FIITJEE

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST – IV

JEE (Advanced)-2021

PAPER –1

TEST DATE: 01-08-2021

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. C

Sol. $\frac{a}{u_x}+\frac{a}{eu_x}+\frac{a}{e^2u_x}=\frac{2u_y}{g}$ …(1)

Horizontal range:

$R=\frac{2.u_xu_y}{g}$ …(2)

2. B

Sol. $N_1 = \mu N_2$ …(1)

$W = \mu N_1 + N_2$ …(2)

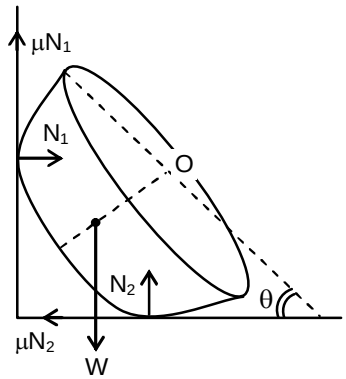
By (1) and (2)

$N_1=\frac{\mu W}{1+\mu^2}$ …(2)

Balancing torque about 'O'

$\mu N_1 \times R + \mu N_2 \times R = W \times \frac{3R}{8}\sin\theta$

$\text{Sin}\,\theta=\frac{8}{3}\left[\frac{\mu+\mu^2}{1+\mu^2}\right]$



3. D

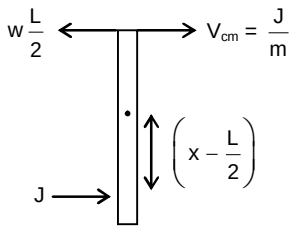
Sol. $\frac{dQ}{dt} = \frac{K(6a^2)(T-T_0)}{x} = n\left(\frac{R}{\gamma-1}\right)\frac{dT}{dt}$

$\int_{T_1}^{T} \frac{dT}{T-T_0} = \frac{6Ka^2(\gamma-1)}{nRx}\int_{o}^{t} dt$

$T = T_0 + (T_1 - T_0)e^{\frac{6Ka^2(\gamma-1)t}{nRx}}$

4. C

Sol. $J\left(x - \frac{L}{2}\right) = I\omega$



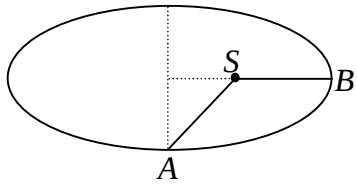
$\omega\frac{L}{2} = V_{cm} = \frac{J}{M}$; $\frac{J\left(x-\frac{L}{2}\right)\frac{L}{2}}{I} = \frac{J}{M}$

$I = \frac{ML^2}{12}$

5. A

Sol. $t_{AB} = \left(\frac{\text{Area SAB}}{\text{Area of ellipse}}\right)\times T$

$= \frac{\left\{\frac{\pi ab}{4} - \frac{1}{2}(b)(ea)\right\}}{\pi ab}\times T = \left(\frac{1}{4} - \frac{e}{2\pi}\right)T$



6. C

Sol. $F = \frac{k}{v}$

$m\frac{dv}{dt} = \frac{k}{v}$; $\int v dv = \frac{k}{m}\int dt$

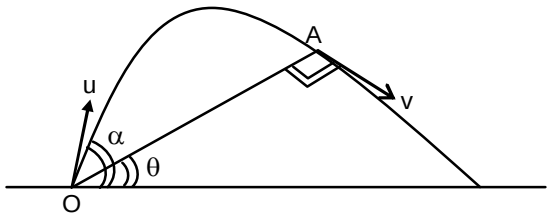
$\frac{mv^2}{2} = kt$

Work done by force = change in kinetic energy.

7. ABD

Sol. Time for OA is

$t = \frac{2u\sin(\alpha-\theta)}{g\cos\theta} = \frac{u\cos(\alpha-\theta)}{g\sin\theta}$ …(1)



Solving this equation we get

$2\tan^2\theta\cos\alpha - \sin\alpha.\tan\theta + \cos\alpha = 0$

This quadratic equation has two roots

$\tan\theta_1 + \tan\theta_2 = \frac{\sin\alpha}{2\cos\alpha} = \frac{1}{2}\tan\alpha$ …(2)

$\text{Tan}\,\theta_1 \times \text{Tan}\,\theta_2 = \dfrac{\text{Cos}\,\alpha}{2\,\text{Cos}\,\alpha} = \dfrac{1}{2}$ …(3)

By (2) & (3)

$$\frac{\text{Tan}\,\theta_1 + \text{Tan}\,\theta_2}{1 - \text{Tan}\,\theta_1 \cdot \text{Tan}\,\theta_2} = \text{Tan}(\theta_1 + \theta_2)$$

$$\frac{\frac{1}{2}\text{Tan}\,\alpha}{1-\frac{1}{2}} = \text{Tan}(\theta_1 + \theta_2)$$

$\theta_1 + \theta_2 = \alpha$

Also, $b^2 \geq 4ac$

$(-\sin\alpha)^2 \geq 4 \times 2\cos\alpha \times \cos\alpha$

Solving we get

$$\alpha \geq \text{Cos}^{-1}\left(\frac{1}{3}\right)$$

8. ABCD

Sol. Apply ampere's law

$$\oint \vec{B} \cdot d\vec{\ell} = \mu_0 i$$

9. ABC

Sol. Loop ABEF

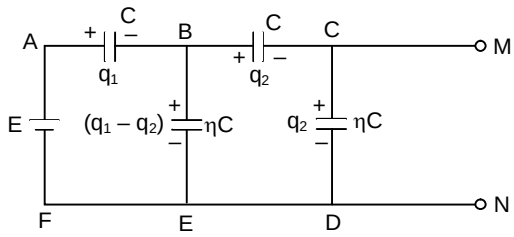
$$\frac{q_1}{c} + \frac{q_1 - q_2}{\eta c} = E \qquad \ldots(1)$$

Loop ABCDEF

$$\frac{q_1}{c} + \frac{q_2}{c} + \frac{q_2}{\eta c} = E \qquad \ldots(2)$$

Bu (1) and (2) we get

$$V_{MN} = \frac{q_2}{\eta c} = \frac{E}{\eta^2 + 3\eta + 1}$$



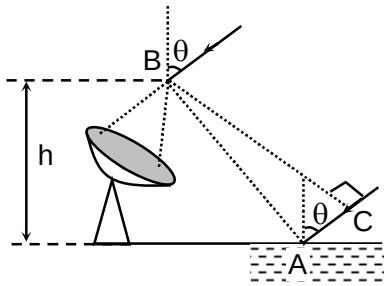
10. AD

Sol. Path difference of the two rays (one coming directly to antenna and other after reflection from water surface

$$\Delta x = AB + AC + \frac{\lambda}{2}$$

$$\Rightarrow \quad \Delta x = \frac{h}{\cos\theta} + \frac{h}{\cos\theta}\cos 2\theta + \frac{\lambda}{2}$$

$$= \quad \frac{h}{\cos\theta}(1 + \cos 2\theta) + \frac{\lambda}{2}$$



or $\Delta x = 2h\cos\theta + \frac{\lambda}{2}$

For constructive interference $\Delta x = n\lambda$

$\Rightarrow \quad 2h\cos\theta + \frac{\lambda}{2} = n\lambda$ or $h = \frac{\left(n - \frac{1}{2}\right)\lambda}{2\cos\theta} = \frac{\lambda}{4\cos\theta}, \frac{3\lambda}{4\cos\theta}, \ldots\ldots$ etc

11. B

Sol. Constant force does not affect T of spring block system.

12. AC

Sol. $\frac{dN_B}{dt} = \lambda_1 N_A - \lambda_2 N_B$

at, $t = t_0$, $\frac{dN_B}{dt} = 0$, $N_B = \frac{3N_0}{2}$

$\Rightarrow \quad 2N_0\lambda_1 e^{-\lambda_1 t_0} - \lambda_0 \frac{3N_0}{2} = 0$

$t_0 = \frac{1}{\lambda_1} \ln \frac{4}{3}\frac{\lambda_1}{\lambda_2}$

SECTION – C

13. 00274.00

Sol. $W = \frac{4}{3}\pi R^3 \rho g \left[h - R + \frac{x}{2}\right] = \frac{4}{3}\pi R^3 \rho g \left[h - R + \frac{\frac{4}{3}\pi R^3}{2A}\right]$

14. 00004.58

Sol. $h = (R - r)[\cos 30° - \cos 60°]$

$mgh = \frac{1}{2}mv^2 + \frac{1}{2}mr^2\omega^2$

$x = \frac{v^2 \sin^2 30^o}{2g}$

15. 00002.72

Sol. Loss in KE = $K_f - K_i$

$K_f = \frac{1}{2}mv_{cm}^2 + \frac{1}{2}I_{cm}\omega^2$

$K_i = \frac{1}{2}(0.08)(10)^2 + \frac{1}{2}(0.08)(6)^2$

Apply conservation of momentum and angular momentum to get V_{cm} and ω.

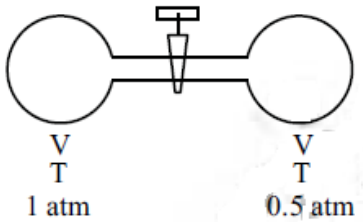
16. 00002.66

Sol. The observed reading will be = 10 mm + 1 × 01 mm = 1.01 cm

$\therefore \text{density} = \frac{\text{Mass}}{\text{vol}} = \frac{2.7369}{(1.01)^3} = 2.65654 \approx 2.66 \text{ gm/cm}^3$

17. 00000.60

Sol. After opening of at equilibrium temperature and pressure of whole gas is T_1 and P_1



$n_1 = \frac{1\times V}{RT}$, $n_2 = \frac{0.5\times V\times 4}{RT}$

$n_1 + n_2 = n$

$\frac{V}{RT} + \frac{V\times 4}{2RT} = \frac{5VP_1}{RT_1}$

$\frac{3V}{RT} = \frac{5VP_1}{RT_1}$; $\frac{P_1}{T_1} = \frac{0.6}{T}$

$\Delta Q = 0$, $\Delta W = 0$

$\therefore\ \Delta U = 0$

$n_1C_VT + n_2C_VT = (n_1 + n_2)C_VT_1$

$T_1 = T$

$\frac{P_1}{T} = \frac{0.6}{T}$

$P_1 = 0.60$ atm

18. 00000.25

Sol. $B\ell v = i_1R$ …(1)

$B\ell v = \frac{q}{c}$ but $\frac{dq}{dt} = i_2$

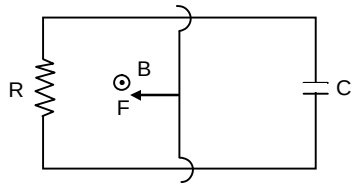
$q = B\ell vc$

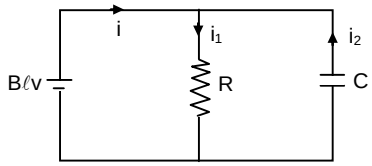
$i_i = B\ell ca$ …(2)

$F - (i_1 + i_2)\, B\ell = ma$

Solving this we can find terminal velocity

$v = \frac{FR}{B^2\ell^2} = \frac{FR}{4FR} = 0.25$





Chemistry

PART – II

SECTION – A

19. D

Sol. $\left(\frac{2.178-2.165}{2.178}\right)=5.96\times10^{-3}$

20. D

Sol. $K_p = P_{CO_2} = 1$

$\log K_p = 10 - 8500 / T$

$\log 1 = 10 - 8500/T$

$T = 850\ K$

21. B

Sol. Normality $= \frac{wt\times n\times1000}{MM\times V}$

$\frac{1}{10}=\frac{w\times2\times1000}{126\times250}$

$w=0.1575$ gm

22. C

Sol. Lindlar's catalyst forms cis-alkene.

23. C

Sol. $(NH_4)_2Cr_2O_7 \xrightarrow{Heat} Cr_2O_3 + N_2 + 4H_2O$

24. D

Sol. $\Delta H = \Delta E + \Delta nRT$

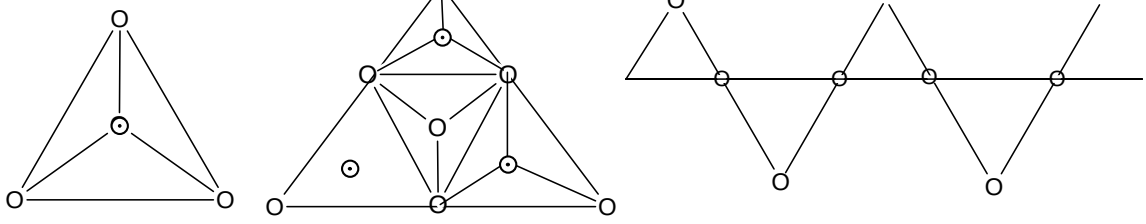
If $\Delta n > 0$ $\quad \Delta H > \Delta E$.

25. ABC

Sol. $LiAlH_4$, converts carbonyl compound into alcohol.

26. ABC

Sol.

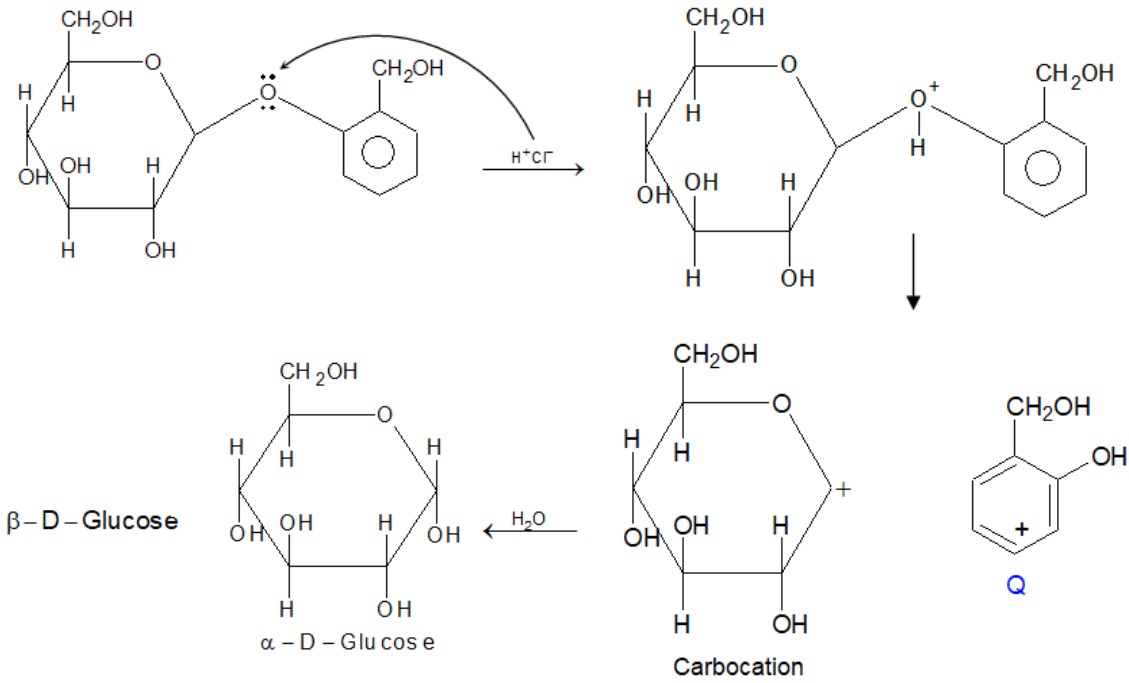


27. ACD

Sol. NO, O_2^{2+} and oxygen having unpaired electrons in molecular orbital.

28. ABCD

Sol.



29. AC

Sol. $pH = \frac{1}{2}PK_w - \frac{1}{2}PK_b - \frac{1}{2}\log c$

Using this we will get conc. & mol mass

30. ACD

Sol. Rate can be changed by changing the concentration of reacting species involved in rate law expression and by the temperature.

SECTION – C

31. 00006.90

Sol. For 1 mole of ideal gas, PV = constant; Isothermal process.

$W_{reversible} = -nRT \ln(V_2/V_1) = -PV \ln(P_1/P_2)$

32. 00002.00

Sol. $\Delta H = E_{a_f} - E_{a_b}$

$E_{a_b} = 120$ kJ mol^{-1}

After addition of catalyst

$\Delta H = -40$ kJ mol^{-1}

$E_{a_f} = 20$ kJ mol^{-1}

$E_{a_b} = 60$ kJ mol^{-1}

33. 00000.50

Sol. PV = nRT

$n = \frac{PV}{RT} = \frac{2 \times 400 \times 10^{-3}}{0.082 \times 100}$ mole

Using this value of mole we can calculate pressure of gas in new cylinder.

PV = nRT

$p = \frac{nRT}{V} = \frac{2 \times 400 \times 10^{-3} \times 0.082 \times 100}{0.082 \times 100 \times 1.6} = 0.5$ atm

34. 00077.07

Sol. $2r = \frac{\sqrt{3}a}{4}$

$r = \frac{\sqrt{3}a}{8}$

35. 00001.50

Sol. $\frac{x}{y+z} = \frac{18}{12} = 1.50$

A is $Na_6P_6O_{18}$

36. 00001.60

Sol. $\frac{x}{5} = \frac{8}{5} = 1.60$

All of them can form.

Mathematics

PART – III

SECTION – A

37. A

Sol. As, $f(x) = x^4\tan(x^3) - x\ln(1+x^2)$ is odd, $\frac{d^4f(x)}{dx^4} = 0$ at $x = 0$.

38. A

Sol. $\lim\limits_{t\to 0}\frac{\int_0^t \sin t^2\,dt}{\frac{1}{2}t\sin t^2} = \lim\limits_{t\to 0}\frac{2\sin t^2}{\sin t^2 + 2t^2\cos t^2} = \lim\limits_{t\to 0}\frac{2}{1+2\frac{t^2}{\tan t^2}} = \frac{2}{1+2} = \frac{2}{3}$

39. B

Sol. On expanding we get $\left(\vec{a}.\vec{c}\right)\vec{a} - \left(\vec{a}.\vec{a}\right)\left(\vec{c}\right) = -\vec{b}$

$\Rightarrow \vec{c} - \left(\vec{a}.\vec{c}\right)\vec{a} = \vec{b}$

$\Rightarrow$ (a) is correct and (b) is ruled out.

$\Rightarrow \left|\vec{c} - \left(\vec{a}.\vec{c}\right)\vec{a}\right|^2 = \left|\vec{b}\right|^2$

If θ is the angle between $\vec{a}$ and $\vec{c}$ then

$4 + \left(2\cos\theta\right)^2 - 2\left(2\cos\theta\right)^2 = 1$

$\Rightarrow \cos\theta = \pm\frac{\sqrt{3}}{2}$

$\Rightarrow \theta = 30^o; 150^o$

40. D

Sol. $g(x) \Rightarrow \begin{cases} -x^2\cot^{-1}x; & x > 0 \\ -x^2\cot^{-1}x; & x < 0 \\ 0\ ; x = 0 \end{cases} \Rightarrow$ twice differentiable

$h(x) = \sin^2 x \rightarrow$ differentiable

$k'(x) = \begin{cases} 4x^3.\cos\frac{1}{x} + x^2.\sin\frac{1}{x}; & x \neq 0 \\ 0; & x = 0 \end{cases}$

twice differentiable

41. B

Sol. Since the system has non – trivial solution, therefore $\begin{vmatrix} a & 1 & 1 \\ 1 & b & 1 \\ 1 & 1 & c \end{vmatrix} = 0$

Applying $R_1 \to R_1 - R_2, R_2 \to R_2 - R_3$, we get $\Delta = \begin{vmatrix} a-1 & 1-b & 0 \\ 0 & b-1 & 1-c \\ 1 & 1 & c \end{vmatrix} = 0$

$\Rightarrow (a-1)(b-1)c-(1-c)+(1-b)(1-c)=0$

$\Rightarrow (1-a)(1-b)c+(1-c)+(1-b)(1-c)=0$

Dividing throughout by $(1-a)(1-b)(1-c)$,

we get $\frac{c}{1-c}+\frac{1}{1-b}+\frac{1}{1-a}=0$

$\Rightarrow -1+\frac{1}{1-c}+\frac{1}{1-b}+\frac{1}{1-a}=0$

$\Rightarrow \frac{1}{1-c}+\frac{1}{1-a}+\frac{1}{1-b}=1.$

42. B

Sol. The volume is invariant under translation.

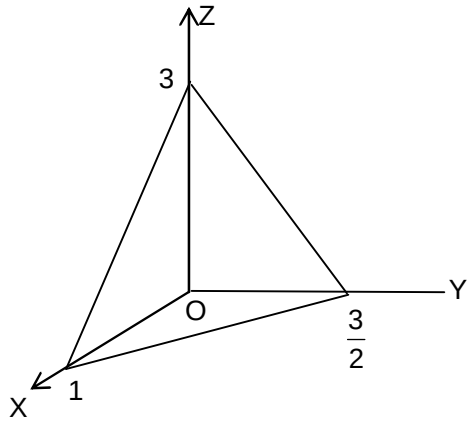
If $X=x-\frac{4}{3}$, $Y=y-\frac{3}{2}$, $Z=z+4$, then the planes become $3|X|+2|Y|+|Z|=3$

The plane $3X+2Y+Z=\pm 3, -3X+2Y+Z=\pm 3.$

$3X+2Y-Z=\pm 3$, $3X+2Y-Z=\pm 3$ form an octahedron.

The plane $3X+2Y+Z=3$ forms a tetrahedron with coordinate planes in the positive octant of volume $\frac{1}{6}.1.\frac{3}{2}.3=\frac{3}{4}$

Total volume $=8\times\frac{3}{4}=6$



43. ABD

Sol. Volume $=\left|\left[2\vec{b}\times\vec{c}\quad 3\vec{c}\times\vec{a}\quad 4\vec{a}\times\vec{b}\right]\right|=18$

$\Rightarrow \quad 24\left[\vec{a}\quad\vec{b}\quad\vec{c}\right]^2=18$

$\Rightarrow \quad \left|\left[\vec{a}\quad\vec{b}\quad\vec{c}\right]\right|=\frac{\sqrt{3}}{2}$

Now, $\left[\vec{a}\quad\vec{b}\quad\vec{c}\right]=\begin{vmatrix}(1+\sin\theta) & \cos\theta & \sin 2\theta\\ \sin\left(\theta+\frac{2\pi}{3}\right) & \cos\left(\theta+\frac{2\pi}{3}\right) & \sin\left(2\theta+\frac{4\pi}{3}\right)\\ \sin\left(\theta-\frac{2\pi}{3}\right) & \cos\left(\theta-\frac{2\pi}{3}\right) & \sin\left(2\theta-\frac{4\pi}{3}\right)\end{vmatrix}$

Applying $R_1\to R_1+R_2+R_3$ and expanding

$\left|\left[\vec{a}\quad\vec{b}\quad\vec{c}\right]\right|=\sqrt{3}\left|\cos 3\theta\right|=\frac{\sqrt{3}}{2}$

$\Rightarrow \cos 3\theta=\pm\frac{1}{2}\Rightarrow 3\theta=\frac{\pi}{3},\frac{2\pi}{3},\frac{4\pi}{3}\Rightarrow\theta=\frac{\pi}{9},\frac{2\pi}{9},\frac{4\pi}{9}$

44. CD

Sol. Clearly $p_1 = \frac{1}{2}$

$p_2 = p(HH) + p(T) = \frac{3}{4}$

Now, the score n can be obtained in two distinct ways.
I. By throwing head when the score is n – 1
II. By throwing tail when the score is n – 2

$\Rightarrow p_n = (p_{n-1})\frac{1}{2} + (p_{n-2})\frac{1}{2} = \frac{1}{2}(p_{n-1} + p_{n-2})$

Also, from this $P_4 = \frac{11}{16}$.

45. ABC

Sol. $A \subseteq A \cup B$

$\Rightarrow P(A) \le P(A \cup B) \Rightarrow P(A \cup B) \ge \frac{3}{4}$

Also,

$$P(A \cap B) = P(A) + P(B) - P(A \cup B)$$
$$\ge P(A) + P(B) - 1$$
$$= \frac{3}{4} + \frac{5}{8} - 1 = \frac{3}{8}$$

Now,

$$A \cap B \subseteq B$$

$\Rightarrow P(A \cap B) \le P(B) = \frac{5}{8}$

$\therefore \quad \frac{3}{8} \le P(A \cap B) = \frac{5}{8}$

and

$$P(A \cap B') = P(A) - P(A \cap B)$$

$\Rightarrow \frac{3}{4} - \frac{5}{8} \le P(A \cap B') \le \frac{3}{4} - \frac{3}{8}$

$\Rightarrow \frac{1}{8} \le P(A \cap B') \le \frac{3}{8}$

$\therefore P(A \cap B) = P(B) = P(A' \cap B)$ [Using Equation (1)]

$\Rightarrow \frac{3}{8} \le P(B) - P(A' \cap B) \le \frac{5}{8}$

$\Rightarrow 0 \le P(A' \cap B) \le \frac{1}{4}$

46. BCD

Sol. $x^4+4x^3-8x^2=-k$

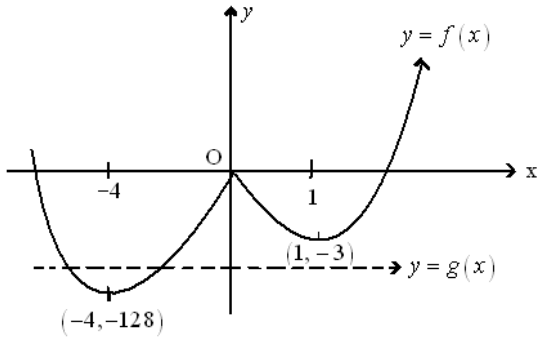
$\Rightarrow \quad f(x)=-k$

Where

$f(x)=x^4+4x^3-8x^2=x^2\left(x^2+4x-8\right)$

Let $g(x)=-k$

From the graph the following cases arise :



1. When $-3\le -k\le 0, \Rightarrow 0\le k\le 3$

   In this case, $y=-k$ intersect at four points.

2. When $-4\le -k<-3, \Rightarrow 3<k\le 4$

   In this case, $y=-k$ intersect at two points, the given equation has two real roots.

3. When $k<0, \Rightarrow -k>0$

   In this case, there are two points of intersection. So, the equation has two real roots.

47. ABCD

Sol. $(gof)(x)=(2x-\sin x)^{1/3}$

Range of (gof) (x) is R.

48. BC

Sol. $\lim\limits_{x\to\infty}\sqrt{x}\left(\sqrt{x+1}-\sqrt{x}\right)$

$=\lim\limits_{x\to\infty}\dfrac{\sqrt{x}}{\sqrt{x+1}+\sqrt{x}}=\lim\limits_{x\to\infty}\dfrac{1}{\sqrt{1+\dfrac{1}{x}}+1}=\dfrac{1}{2}$

Option (A): $\lim\limits_{x\to0}\dfrac{\ln(1+x)-x}{x^2}=\lim\limits_{x\to0}\dfrac{\left(x-\dfrac{x^2}{2}+\dfrac{x^3}{3}-\ldots\right)-x}{x^2}=-\dfrac{1}{2}$

Option (B): $\lim\limits_{x\to0}\dfrac{1-\cos x}{x^2}=\lim\limits_{x\to0}\dfrac{2\sin^2\dfrac{x}{2}}{4\times\left(\dfrac{x}{2}\right)^2}=\dfrac{1}{2}$

Option (C): $\lim\limits_{x\to0}\dfrac{\sqrt{1+x}-1}{x}=\lim\limits_{x\to0}\dfrac{(1+x)^{1/2}-(1)^{1/2}}{(1+x)-1}=\dfrac{1}{2}$

Option (D): $\lim\limits_{x\to0}\dfrac{\sqrt{x}}{\sqrt{x+\sqrt{x^2+2x}}}=\lim\limits_{x\to0}\dfrac{1}{\sqrt{1+\sqrt{1+\dfrac{2}{x}}}}=\dfrac{1}{\sqrt{2}}$

SECTION – C

49. 00006.00

Sol. Let $z = x_1 + iy_1, w = x_2 + iy_2$

$\therefore |z-3| = \text{Re}\, z \Rightarrow (x_1-3)^2 + y_1^2 = x_1^2$ ....(i)

$|w-3| = \text{Re}\, w \Rightarrow (x_2-3)^2 + y_2^2 = x_2^2$ ...(ii)

$\arg(z-w) = \frac{\pi}{4} \Rightarrow \frac{y_1-y_2}{x_1-x_2} = \tan\frac{\pi}{4} = 1$ ...(iii)

By solving (i), (ii) and (iii) we get

$\Rightarrow \frac{y_1-y_2}{x_1-x_2} = \frac{6}{y_1+y_2}$

From (iii) $y_1 + y_2 = 6$

50. 00002.50

Sol. Let $\vec{r} = \lambda\vec{b} + \mu\vec{c}$ and $\vec{c} = \pm(xi + yj)$. Since $\vec{c}$ and $\vec{b}$ are perpendicular, we have

$4x + 3y = 0 \quad \Rightarrow \quad \vec{c} = \pm x\left(i - \frac{4}{3}j\right)$

$\pm 1 =$ proj. of $\vec{r}$ on $\vec{b} = \frac{\vec{r}.\vec{b}}{|\vec{b}|} = \frac{(\lambda\vec{b} + \mu\vec{c}).\vec{b}}{|\vec{b}|} = \frac{\lambda\vec{b}.\vec{b}}{|\vec{b}|}$

$\left[\because \vec{b}.\vec{c} = 0\right]$

$= \lambda|\vec{b}| = 5\lambda$. Hence $\lambda = \pm\frac{1}{5}$

Also, $\pm 2 = $ proj. of $\vec{r}$ on $\vec{c} = \frac{\vec{r}.\vec{c}}{|\vec{c}|}$

$= \frac{(\lambda\vec{b} + \mu\vec{c}).\vec{c}}{|\vec{c}|} = \mu|\vec{c}| = \frac{5}{3}\mu x$

Thus, $\mu x = \pm\frac{6}{5}$. Therefore,

$\vec{r} = \frac{1}{5}(4i + 3j) + \frac{6}{5}\left(i - \frac{4}{3}j\right) = \pm(2i - j)$

$\vec{r} = \frac{1}{5}(4i + 3j) - \frac{6}{5}\left(i - \frac{4}{3}j\right) = \pm\left(-\frac{2}{3}i + \frac{11}{5}j\right)$

Thus there are four such vectors

$\sum_{i=1}^{4}|\vec{r_i}|^2 = 2|2i - j|^2 + 2\left|-\frac{2}{5}i + \frac{11}{5}j\right|^2 = 20$

51. 00000.50

Sol. $a(x^2 - x + 1) - (x^2 + x + 1) \geq 0$

$\Rightarrow a \geq \frac{x^2+x+1}{x^2-x+1}$ or $a \geq \frac{\left(x^2-x+1\right)+2x}{x^2-x+1}$

$\Rightarrow a \geq 1+\frac{2x}{x^2-x+1}$ or $a \geq 1+\frac{2}{\left(x+\frac{1}{x}\right)-1}$

Now, for $x \geq 2$, the minimum value of $x+\frac{1}{x}$ is $\frac{5}{2}$.

$\Rightarrow a \geq \frac{7}{3}$

52. 00007.00

Sol. $x_1+x_2+x_3=12$

$t_1+t_2+t_3=9$ Total solutions $\left(t_i=x_i-1\right)$

$y_1+t_2+t_3=3$ Extra solutions $\left(y_1=t_1-6\right)$

53. 00030.25

Sol. Let $\alpha-d, \alpha, \alpha+d$ are roots

$\therefore 3\alpha=-a$ (1)

$3\alpha^2-d^2=b$ (2)

$\alpha\left(\alpha^2-d^2\right)=\frac{1}{9}$ (3)

From (1) and (2), $d^2=\frac{a^2}{3}-b$

Put d^2 in (3) to get

$2a^3-9ab+1=4$

54. 00000.50

Sol. $9A^2-6A+I=\begin{bmatrix}6 & 27\\ -10 & -1\end{bmatrix}B^{-1}$

$\Rightarrow (3A-I)^2=81$

$\Rightarrow |3A-I|=\pm 9$ (1)

Let $A=\begin{bmatrix}a & b\\ c & d\end{bmatrix}$

$\therefore g(ad-bc)=18$ or 0 (using equation 1)

Since $|A| \neq 0$

$\therefore ad-bc=2 \Rightarrow |A|=2$